

EGCO 425 – Chapter 1

Introduction to data mining

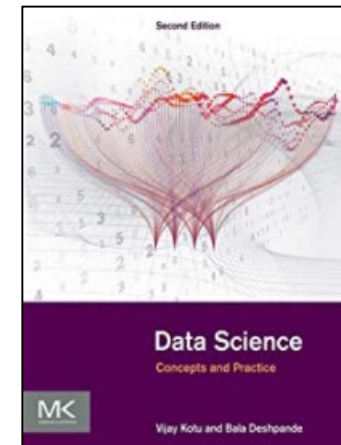
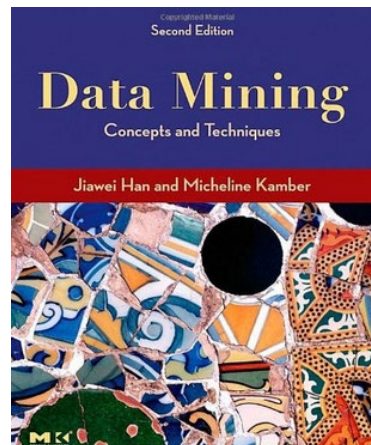
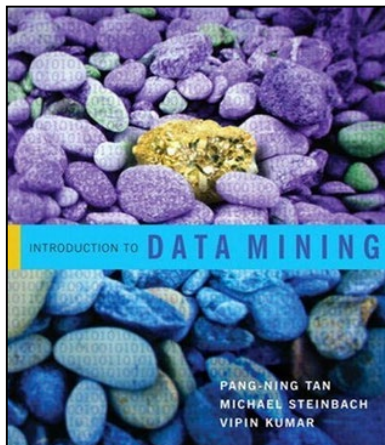
Data mining tools (RapidMiner)

References

- [1] Tan et al., Introduction to Data Mining (Chapter 1)
- [2] Han & Kamber, Data Mining: Concepts and Techniques (Chapter 1)
- [3] Kotu & Deshpande, Data Science: Concepts and Practice (Chapters 1-2, 15)
- [4] RapidMiner Doc, <https://docs.rapidminer.com/latest/studio/operators/>

Textbooks

- ✦ [Tan, Steinbach & Kumar](#). Introduction to Data Mining. Pearson Addison Wesley
- ✦ [Han & Kamber](#). Data Mining: Concepts & Techniques. Morgan Kaufmann
- ✦ [Kotu & Deshpande](#). Data Science: Concepts and Practice. Morgan Kaufmann



Data Mining and Data Science

- ⊕ Data science = a field of study that focuses on the science of data
 - ▶ Including big data, data mining, data visualization, mathematics, statistics, business intelligence, etc.
- ⊕ Data mining
 - ▶ Process oriented
 - ▶ Emphasize on techniques (machine learning + statistics) to extract **hidden patterns** from a **large amount of structured data** e.g. tables, transactions, relational data
 - *Other areas that handle other types of data are, for example, text mining (unstructured data), social network mining (graph data)*
 - ▶ Data mining activities are essential parts of data science pipeline

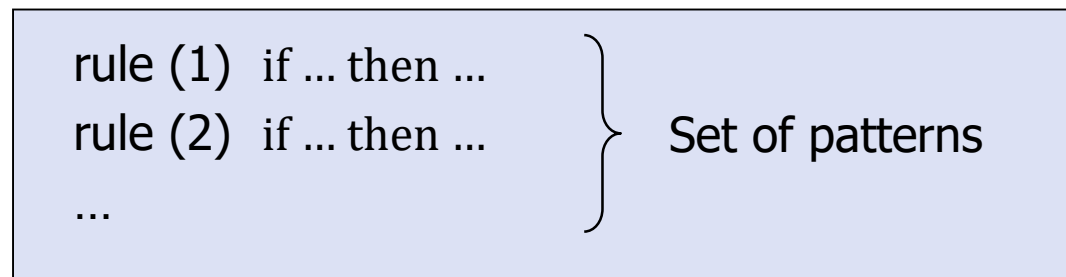
What is and is not data mining ?

No	Yes
Search particular information , e.g. jjapaguri (Korean noodle), on Google	Group similar recipes based on their ingredients, cooking instructions, etc.
Analyze the record of a patient with a certain disease	Find a way to group patients with the same disease , based on a certain number of specific symptoms
Analyze figures in a financial report of a trade company	Use trade company's data concerning sales to identify the customer main profiles
<u>In summary</u> Direct search/calculation/analysis on a particular object	<u>In summary</u> Extract patterns/models/etc. from a large amount of data

Extracted Knowledge

- ⊕ Model = global structure summarizing the whole data set
- ⊕ Pattern = statement concerning relationship between certain variables with certain values
 - ▶ Can be expressed as descriptions, rules, trees, etc.
 - ▶ Structured pattern : can be examined, reasoned about, and used to inform future decisions
 - ▶ Unstructured pattern : a black box & incomprehensible

A model



Related Fields

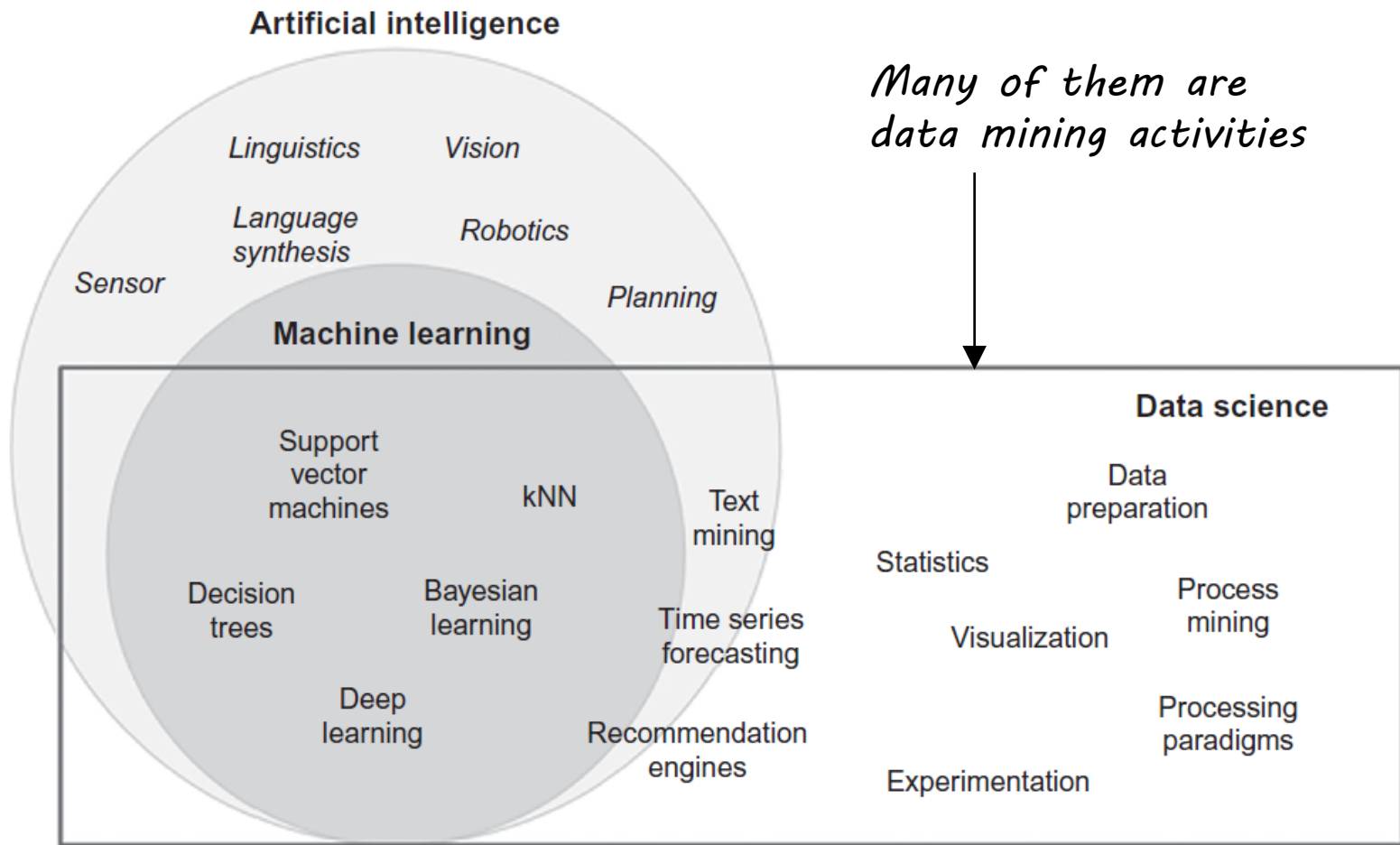


FIGURE 1.1

Artificial intelligence, machine learning, and data science.

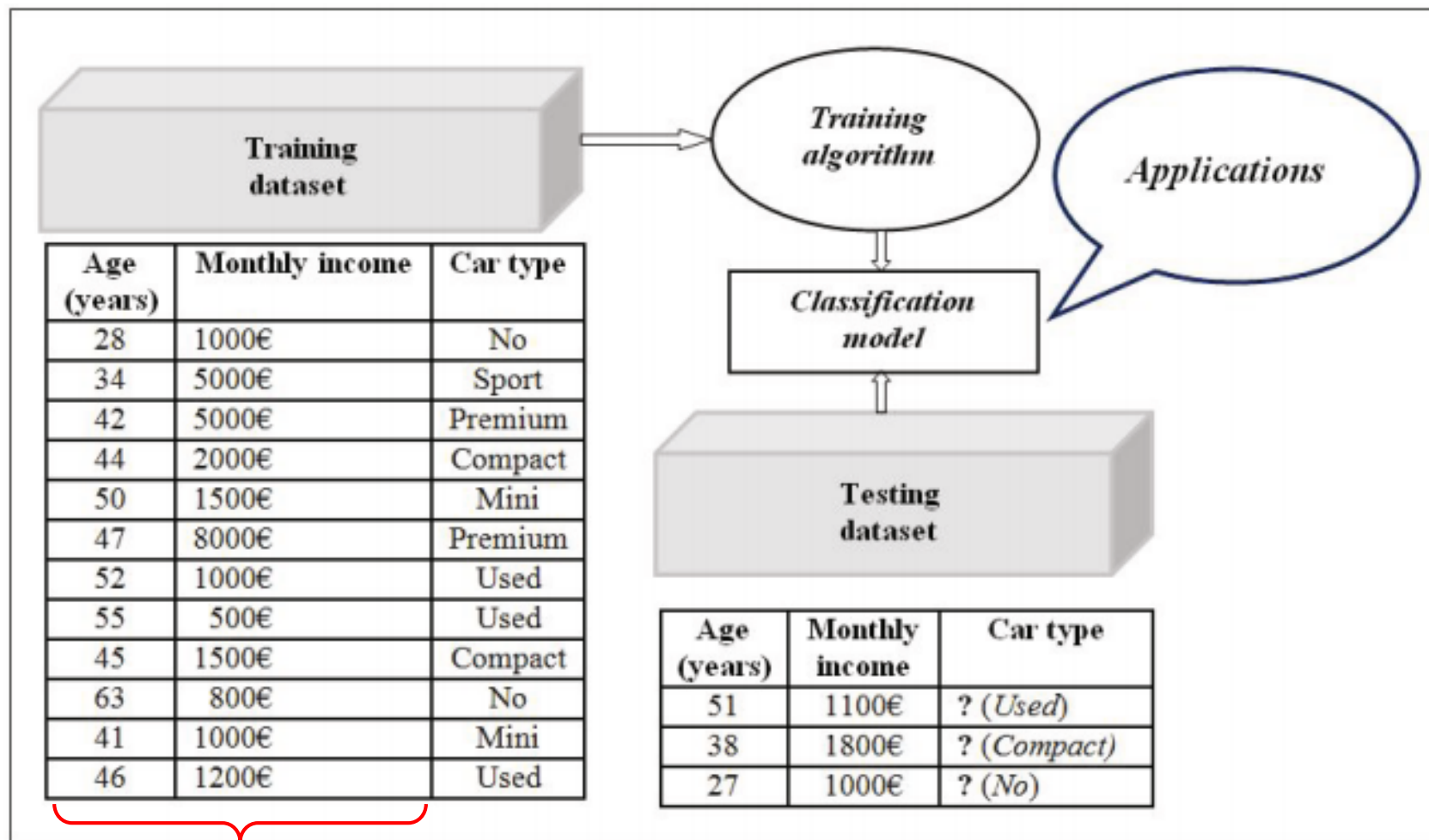
Relationship with AI, Machine Learning

⊕ Artificial intelligence (AI)

- ▶ Giving machines the capability of mimicking human behavior, particularly cognitive functions
- ▶ Learning is an important part of human capability

⊕ Machine learning (ML)

- ▶ Sub-field or tool in AI
- ▶ Providing machines with the capability of learning from experience
 - Experience = training data (independent variables or attributes + class labels)
 - Output = model of relationship between attributes to predict class
 - Usually refer to prediction tasks



Attributes

Class

Data Mining Tasks

⊕ Predictive or supervised learning

- ▶ Construct a model of independent variables to predict dependent or target variable
- ▶ Target variable is specified (i.e. supervised) by user
- ▶ Main tasks
 - Classification predict class labels
 - Regression predict numeric values
- ▶ Many algorithms are complicated, but evaluation is straightforward as we measure whether the prediction is correct or not

⊕ Descriptive or unsupervised learning

- ▶ Describe patterns or trends lying within the data
- ▶ Create models without user-defined configuration
- ▶ All variables are independent
- ▶ Main tasks
 - Clustering group similar objects
 - Association find strongly associated items
- ▶ Algorithms are simple, but evaluation is tricky & depends on different evaluation criteria

Tasks	Description	Algorithms	Examples
Classification	Predict if a data point belongs to one of the predefined classes. The prediction will be based on learning from a known dataset	Decision trees, neural networks, Bayesian models, induction rules, <i>k</i> -nearest neighbors	Assigning voters into known buckets by political parties, e.g., soccer moms Bucketing new customers into one of the known customer groups
Regression	Predict the numeric target label of a data point. The prediction will be based on learning from a known dataset	Linear regression, logistic regression	Predicting the unemployment rate for the next year Estimating insurance premium
Anomaly detection	Predict if a data point is an outlier compared to other data points in the dataset	Distance-based, density-based, LOF	Detecting fraudulent credit card transactions and network intrusion
Time series forecasting	Predict the value of the target variable for a future timeframe based on historical values	Exponential smoothing, ARIMA, regression	Sales forecasting, production forecasting, virtually any growth phenomenon that needs to be extrapolated
Clustering	Identify natural clusters within the dataset based on inherent properties within the dataset	<i>k</i> -Means, density-based clustering (e.g., DBSCAN)	Finding customer segments in a company based on transaction, web, and customer call data
Association analysis	Identify relationships within an item set based on transaction data	FP-growth algorithm, a priori algorithm	Finding cross-selling opportunities for a retailer based on transaction purchase history
Recommendation engines	Predict the preference of an item for a user	Collaborative filtering, content-based filtering, hybrid recommenders	Finding the top recommended movies for a user
LOF, <i>local outlier factor</i> ; ARIMA, <i>autoregressive integrated moving average</i> ; DBSCAN, <i>density-based spatial clustering of applications with noise</i> ; FP, <i>frequent pattern</i> .			

Top 10 Data Mining Algorithms

⌘ By IEEE International Conference on Data Mining (ICDM), 2006

- | | | |
|-------------------------------------|-------------|-----------------------------|
| ☒ | C4.5 | classification |
| ☒ | K-Means | clustering |
| ☒ | SVM | classification |
| ☒ | Apriori | association |
| ☒ | EM | clustering |
| ☐ | PageRank | information retrieval |
| ☒ | AdaBoost | ensemble classification |
| ☒ | KNN | classification |
| ☒ | Naïve Bayes | classification |
| ☒ | CART | classification + regression |

Choosing data mining approach/method

- ⊕ Variables to use in the analysis & their constraints
- ⊕ Data
 - ▶ Reliable source, data collection
 - ▶ Volume
 - ▶ Potential errors
- ⊕ Natures of candidate methods
 - ▶ How each algorithm works
 - ▶ Constraints
 - ▶ Parameters
- ⊕ Evaluation criteria
 - ▶ May depend on application domains

Mining Complex Data

⊕ Multimedia data

- ▶ Images, video, audio
- ▶ E.g. content-based retrieval (CBR)

⊕ Spatial data

- ▶ Space-related data such as maps, medical image, chip layout
- ▶ Contain topological & distance information
- ▶ Multidimensional structure

⊕ Temporal data

- ▶ Time-series data (time constraint), sequence data (order constraint)
- ▶ E.g. trend analysis, sequential pattern analysis

⊕ Web mining

- ▶ Graph data, text & images
- ▶ Challenges : amount of data, complexity, dynamic changes
- ▶ E.g. content mining, structure or linkage mining, web usage mining

⊕ Social network mining

- ▶ Graph data, text & images
- ▶ E.g. centrality analysis, community detection, information diffusion, content recommendation

Popular Tools

Data Science Tools

Big data handling

- SQL
- MongoDB
- Neo4j

Data wrangling

- Apache Spark, Apache Hadoop
- Microsoft Excel
- SAS
- Python
- Power BI
- Alteryx
- Trifacta

Data visualization

- Tableau
- QlikView
- D3
- Ggplot2
- FineReport

Data Collection

- Internal and external data sources
- Case studies and interviews
- Surveys, feedbacks, questionnaires

Data transformation

- SSIS
- Oracle Data Integrator
- SAS Integration studio
- SAP data integrator
- Matillion
- Informatica Power Center

Exploratory Data Analysis

- R (R Studio)
- Python
- SAS
- KNIME
- Microsoft Excel

Machine learning

- Python, R
- Weka
- RapidMiner
- H2o
- MATLAB
- DataRobot
- TensorFlow

<https://www.techgeekbuzz.com/data-science-tools/>

Tools (open-source)

✦ **Weka** (<https://www.cs.waikato.ac.nz/ml/weka/>)

- ▶ **W**aikato **E**nvironment for **K**nowledge **A**nalysis
- ▶ Developed by University of Waikato (New Zealand)
- ▶ Java version was released in 1997
- ▶ Popular in academic/research

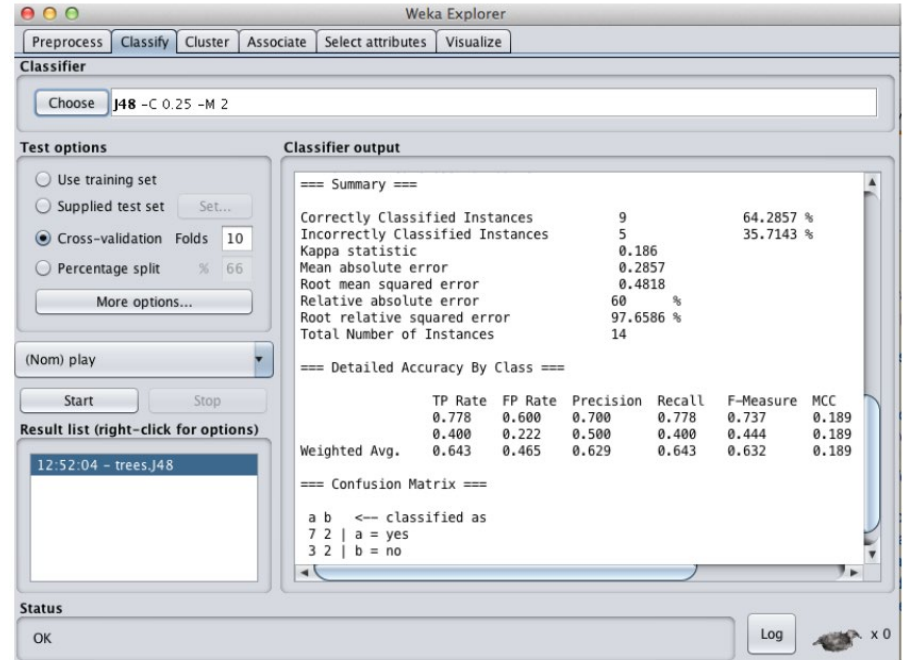


Simple GUI

Data → algorithm → result

Many machine learning methods
are available

Visualization & knowledge flow
are not as good as RapidMiner's



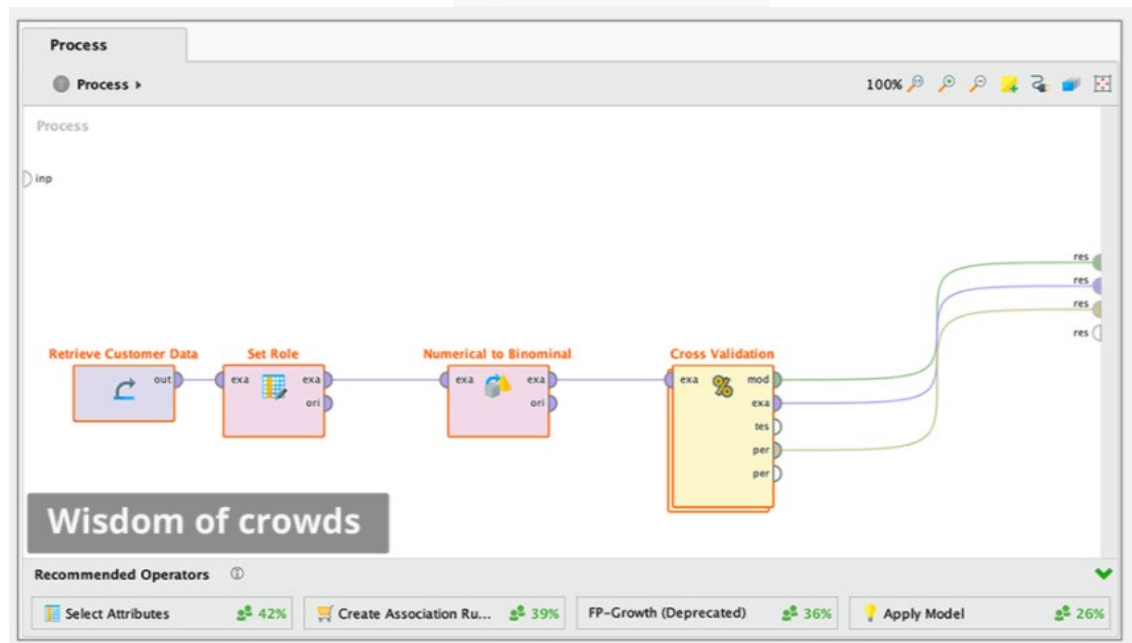
✦ **RapidMiner** (<https://rapidminer.com/products/studio/>)

- ▶ Formerly known as YALE (Yet Another Learning Environment)
- ▶ Written in Java (first released in 2006), by RapidMiner (Germany)
- ▶ Both commercial and free editions
- ▶ Popular in business & research



Use drag & drop GUI
to create and execute
workflows

Templates are available
e.g. direct marketing,
churn analysis

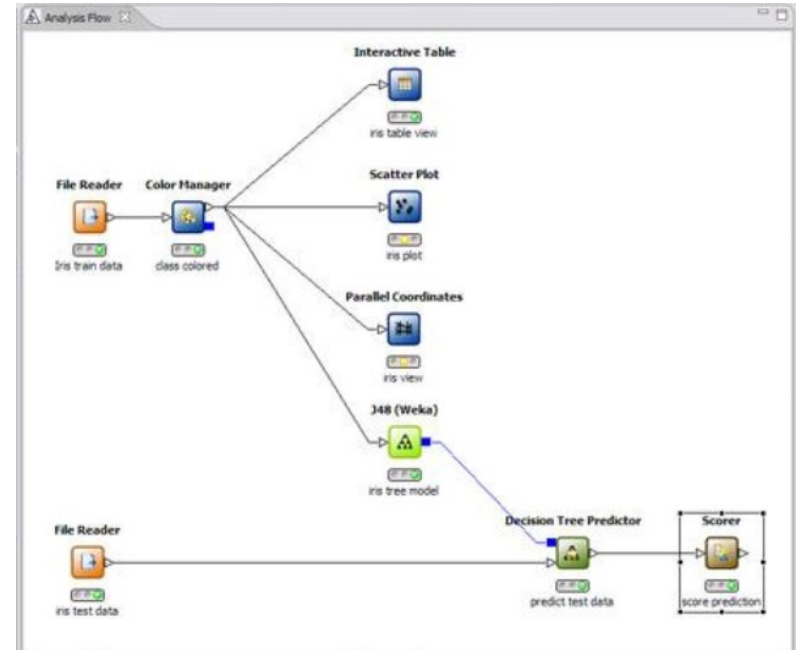


✦ **KNIME** (<https://www.knime.com/>)

- ▶ Konstanz Information Miner
- ▶ Written in Java (first released in 2006) by University of Konstanz (Germany) & now maintained by KNIME (Switzerland)
- ▶ Initially used in pharmaceutical research

Use drag & drop GUI to create and execute workflows

Smaller community than RapidMiner & Weka



Tools (programming languages)

⌘ **R** (<https://www.r-project.org/>)

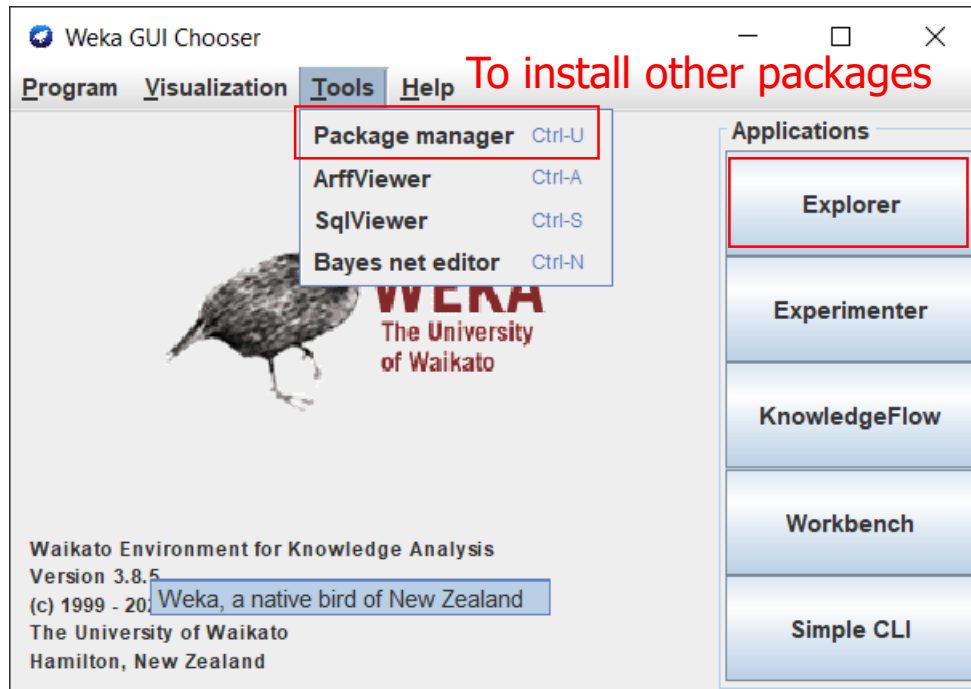
- ▶ Language and environment for statistical computing and graphics
- ▶ Popular IDEs (integrated development environments = coding tools)
 - Rstudio
- ▶ Libraries & packages available in CRAN repository
 - Dplyr, tidyr data manipulation
 - Ggplot2 data visualization
 - Caret, mlr machine learning
 - Knitr report creation

⌘ **Python** (<https://www.python.org/>)

- ▶ General-purpose programming language
- ▶ Popular IDEs
 - Jupyter Notebook, Jupyter Lab
- ▶ Libraries & packages available in PyPI repository
 - Numpy foundation for scientific computing
 - Scipy math & numerical analysis
 - Pandas data manipulation
 - Scikit-learn, tensorflow machine learning
 - NLTK natural language processing
 - Matplotlib, seaborn visualization

Weka Installation

- ✦ Download Weka 3.8 (stable version)
 - ▶ Already bundled with Azul OpenJDK Java VM 11



To install other packages

Quick test of data mining methods

Weka Explorer

Preprocess | Classify | Cluster | Associate | Select attributes | Visualize

Open file... | Open URL... | Open DB... | Generate... | Undo | Edit... | Save...

Filter: Choose None [Apply] [Stop]

Current relation
Relation: weather.symbolic
Instances: 14
Attributes: 5
Sum of weights: 14

Attributes
[All] [None] [Invert] [Pattern]

No.	Name
1	outlook
2	temperature
3	humidity
4	windy
5	play

[Remove]

Selected attribute
Name: outlook
Missing: 0 (0%)
Distinct: 3
Type: Nominal
Unique: 0 (0%)

No.	Label	Count	Weight
1	sunny	5	5.0
2	overcast	4	4.0
3	rainy	5	5.0

Class: play (Nom) [Visualize All]

Status: OK [Log] x 0

ARFF viewer to view data

Can load file.arff, file.csv, file.json

Attribute – Relation File Format (ARFF)

- ⊕ File format used in Weka. Many popular tools also support ARFF
 - ▶ @relation declaration of data set's name
 - ▶ @attribute declaration of attributes
 - ▶ @data raw data
- ⊕ Attribute type → nominal, numeric / real, string, date
- ⊕ Comment start with "%"
- ⊕ Raw data must be in comma-separated vector (CSV) form
 - ▶ Each line = 1 record, ending with a new line character
 - ▶ Within a record, attribute values are separated by comma
 - ▶ Missing data is replaced by "?"


```

weather.nominal - Notepad
File Edit Format View Help
@relation weather.symbolic

@attribute outlook {sunny, overcast, rainy}
@attribute temperature {hot, mild, cool}
@attribute humidity {high, normal}
@attribute windy {TRUE, FALSE}
@attribute play {yes, no}

@data
sunny,hot,high,FALSE,no
sunny,hot,high,TRUE,no
overcast,hot,high,FALSE,yes
rainy,mild,high,FALSE,yes
rainy,cool,normal,FALSE,yes
rainy,cool,normal,TRUE,no
overcast,cool,normal,TRUE,yes
sunny,mild,high,FALSE,no
sunny,cool,normal,FALSE,yes
rainy,mild,normal,FALSE,yes
sunny,mild,normal,TRUE,yes

```

For nominal attribute, all possible values must be given in {value 1, value 2, ...}

Viewer

Relation: **weather.symbolic**

No.	1: outlook Nominal	2: temperature Nominal	3: humidity Nominal	4: windy Nominal	5: play Nominal
1	sunny	hot	high	FALSE	no
2	sunny	hot	high	TRUE	no
3	overcast	hot	high	FALSE	yes
4	rainy	mild	high	FALSE	yes
5	rainy	cool	normal	FALSE	yes
6	rainy	cool	normal	TRUE	no
7	overcast	cool	normal	TRUE	yes
8	sunny	mild	high	FALSE	no
9	sunny	cool	normal	FALSE	yes
10	rainy	mild	normal	FALSE	yes
11	sunny	mild	normal	TRUE	yes
12	overcast	mild	high	TRUE	yes
13	overcast	hot	normal	FALSE	yes
14	rainy	mild	high	TRUE	no

 **Viewer** ✕

Relation: anotherweather

No.	1: outlook Nominal	2: temperature Numeric	3: description String	4: today Date	5: play Nominal
1	sunny	85.0	nice and warm	2015-05-01T12:00:00	no
2	sunny	80.0	nice but rather cold	2015-04-20T15:30:00	no
3	overcast	83.0	quite dull		yes
4	rainy	70.0		2015-07-07T18:00:00	yes

 another - Notepad — □ ✕

File Edit Format View Help

```
@relation anotherweather

@attribute outlook      {sunny, overcast, rainy}
@attribute temperature  numeric
@attribute description  string
@attribute today        date          %% yyyy-mm-ddThh:mm:ss format
@attribute play         {yes, no}

@data
sunny, 85, "nice and warm", 2015-05-01T12:00:00, no
sunny, 80, "nice but rather cold", 2015-04-20T15:30:00, no
overcast, 83, "quite dull", ?,yes
rainy, 70, ?, 2015-07-07T18:00:00, yes
```

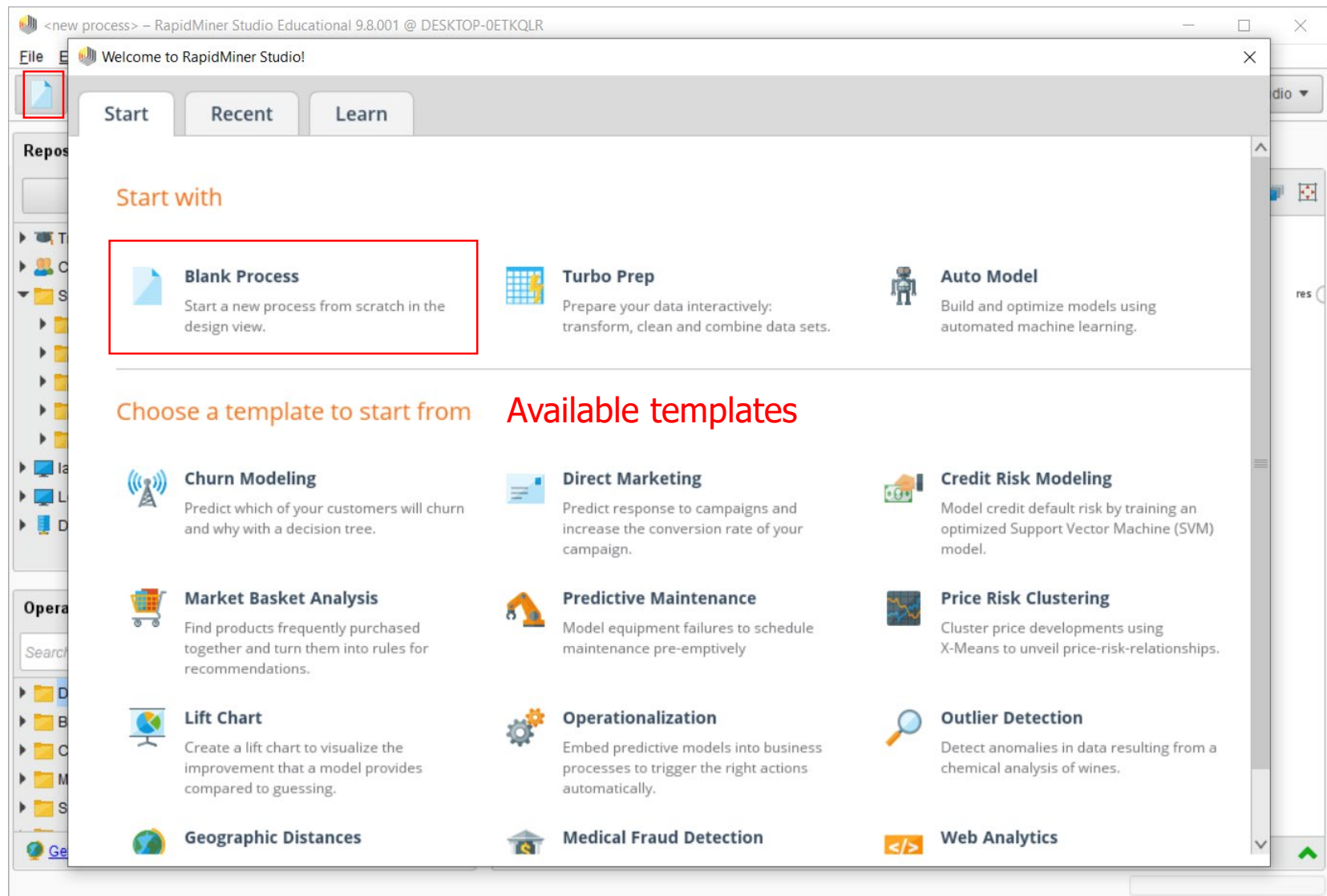
RapidMiner Installation

- ⌘ RapidMiner is now known as Altair AI Studio, which is recently acquired by Siemens
- ⌘ Download AI Studio (2025)
 - ▶ Create account & sign in to download
 - ▶ Java JDK → should already be included in RapidMiner bundle
Otherwise, download it from
(<https://www.oracle.com/java/technologies/downloads/#java21>)
- ⌘ License
 - ▶ Educational license → free, 1 year (renewable), unlimited data rows
 - ▶ Free license → free (no expiration), up to 10K data rows, but some operators such as cluster visualization are not supported

⊕ First run

- ▶ Must login to activate license and complete the installation
- ▶ Activate Wisdom of Crowds (optional) for online recommendation
- ▶ Join Thai RapidMiner community (optional)

Getting start



Churn Modeling Template

<new process> – RapidMiner Studio Educational 9.8.001 @ DESKTOP-0ETKQLR

File Edit Process View Connections Settings Extensions Help

Views: Design Results Turbo Prep Auto Model More Find data, operators...etc All Studio

Repository

Import Data

- Training Resources (conne
- Community Samples (conn
- Samples
- labs (Local)
- Local Repository (Local)
- DB (Legacy)

Parameters

Operators

Search for Operators

- Data Access (58)
- Blending (82)
- Cleansing (28)
- Modeling (167)
- Scoring (14)

[Get more operators from the Marketplace](#)

Process

Process

CHURN MODELING

Why do Telecommunication customers churn? Create a churn model based on observed past churn behavior: Train, optimize and evaluate a decision tree model using a balanced training data set.

Step 1:
Load a customer dataset that contains customer attributes like:

- Age
- Technology used (4G, fiber, etc.)
- Date since he/she is a customer
- Average bill last year
- Number of support calls
- Did he/she abandon last year?

Step 2:
Edit, transform & learn (ETL) and prepare data: Mark the target label column (i.e. the churn indicator) and convert the numerical churn column to binary.

Step 3:
Model validation is key! This cross-validation splits the dataset for training and, then, for independent testing. This splitting is done several times to get a better performance estimate.

Double-click on the operator to take a look at the training itself.

Retrieve Customer ...

Set Role

Numerical to Binomi...

Cross Validation

Outputs:

- A tree model (trained on the complete input data) that analyzes churn behavior and can be applied to any individual customer to estimate churn probability.
- The original input data
- The estimated (i.e. cross-validated) performance of the model.

Recommended Operators

Creating new process

The screenshot displays the RapidMiner Studio Educational 9.8.001 interface. The top menu bar includes File, Edit, Process, View, Connections, Settings, Extensions, and Help. The toolbar features icons for file operations and a 'Run' button (a blue play icon) which is highlighted with a red box. The 'Views' section shows 'Design' and 'Results' tabs, with 'Results' also highlighted by a red box. A red arrow points from the 'Run' button to the 'Results' tab. The 'Repository' panel on the left shows a tree structure with 'Training Resources', 'Community Samples', and 'Samples' folders. The 'Samples' folder is expanded, showing 'data', 'processes', 'Templates', 'Time Series', and 'Tutorials'. A red box highlights the 'labs (Local)' and 'Local Repository (Local)' items. A context menu is open over the 'Local Repository (Local)' item, showing options like 'Create / Connect to Repository', 'Connect to a Project', 'Create subfolder', 'Refresh folder' (F5), 'Sort by', and 'Show location of current process'. The 'Process' panel on the right shows a workflow with a 'Retrieve Titanic Trai...' operator. A red box highlights the 'res' port of this operator. A red arrow points from the 'res' port to the 'Results' tab. The 'Operators' panel at the bottom left shows a list of operators: Data Access (58), Blending (82), Cleansing (28), Modeling (167), Scoring (14), and Validation (30). A red box highlights the 'Data Access (58)' category. A red arrow points from this box to the text 'Drag operators to process panel & connect them to build data analysis process'. The 'Parameters' panel is also visible. The 'Recommended Operators' section at the bottom right shows a list of operators.

Run

Keep data & process
(Local Repository is in
Documents/RapidMiner)

Drag operators to process panel & connect
them to build data analysis process

Right-click at the
operator to add
this note

Drag data from
Samples to here

Out port is connected to res port in order to
complete the workflow (res = show results in
Results tab)

Run the process and check Results tab

Example (1)

Process

Process ▸

Process

Retrieve Titanic Trai...

inp out res res

Right-click at the operator to add this note

Drag data from Samples to here

Out port is connected to res port in order to complete the workflow (res = show results in Results tab)

Run the process and check Results tab

Process can be saved to file.rmp (XML format)

Result History

ExampleSet (Retrieve Titanic Training) ×

Open in Turbo Prep Auto Model

Data

Statistics

Visualizations

Annotations

Row No.	Survived	Age	Passenger ...	Sex	No of Sibling...	No of Parent...	Passenger F...
1	Yes	29	First	Female	0	0	211.338
2	No	2	First	Female	1	2	151.550
3	No	30	First	Male	1	2	151.550
4	No	25	First	Female	1	2	151.550
5	Yes	48	First	Male	0	0	26.550
6	Yes	63	First	Female	1	0	77.958
7	No	39	First	Male	0	0	0
8	Yes	18	First	Female	1	0	227.525
9	Yes	26	First	Female	0	0	78.850
10	Yes	80	First	Male	0	0	30

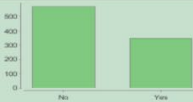
A table (= ExampleSet in RapidMiner)

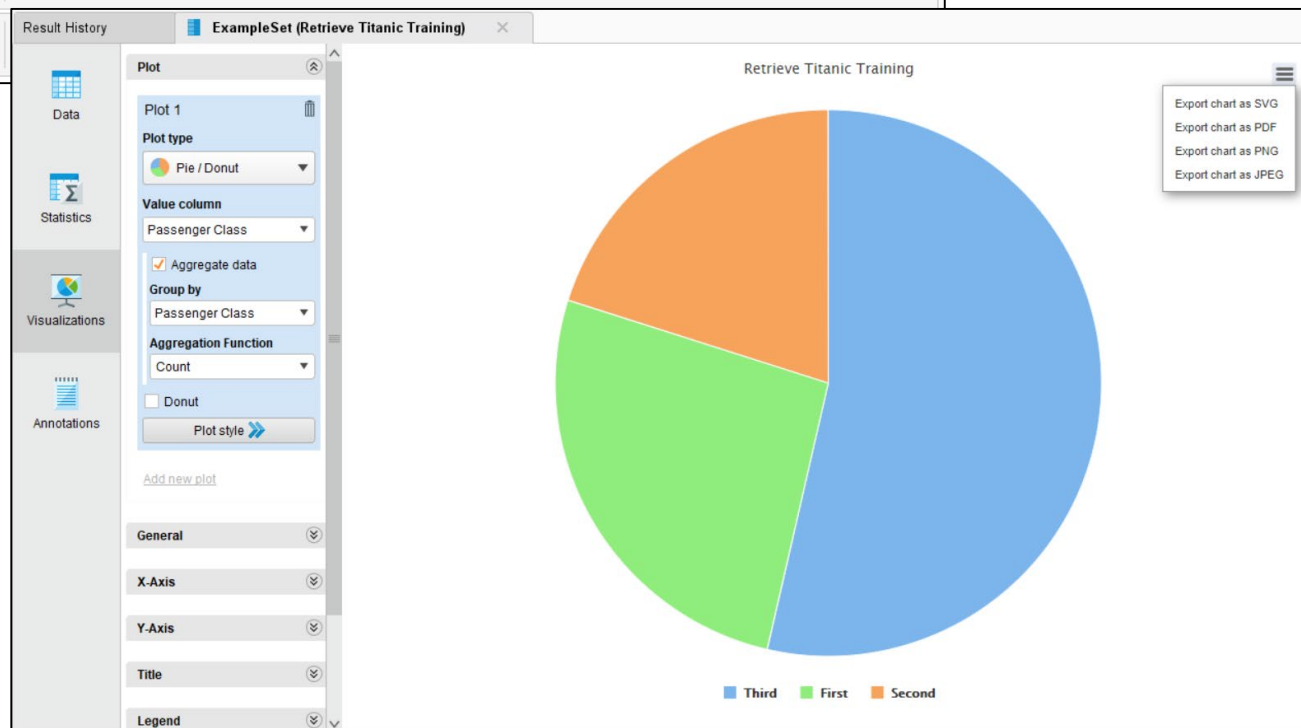
An attribute (= variable)

A record
(= example
in RapidMiner)

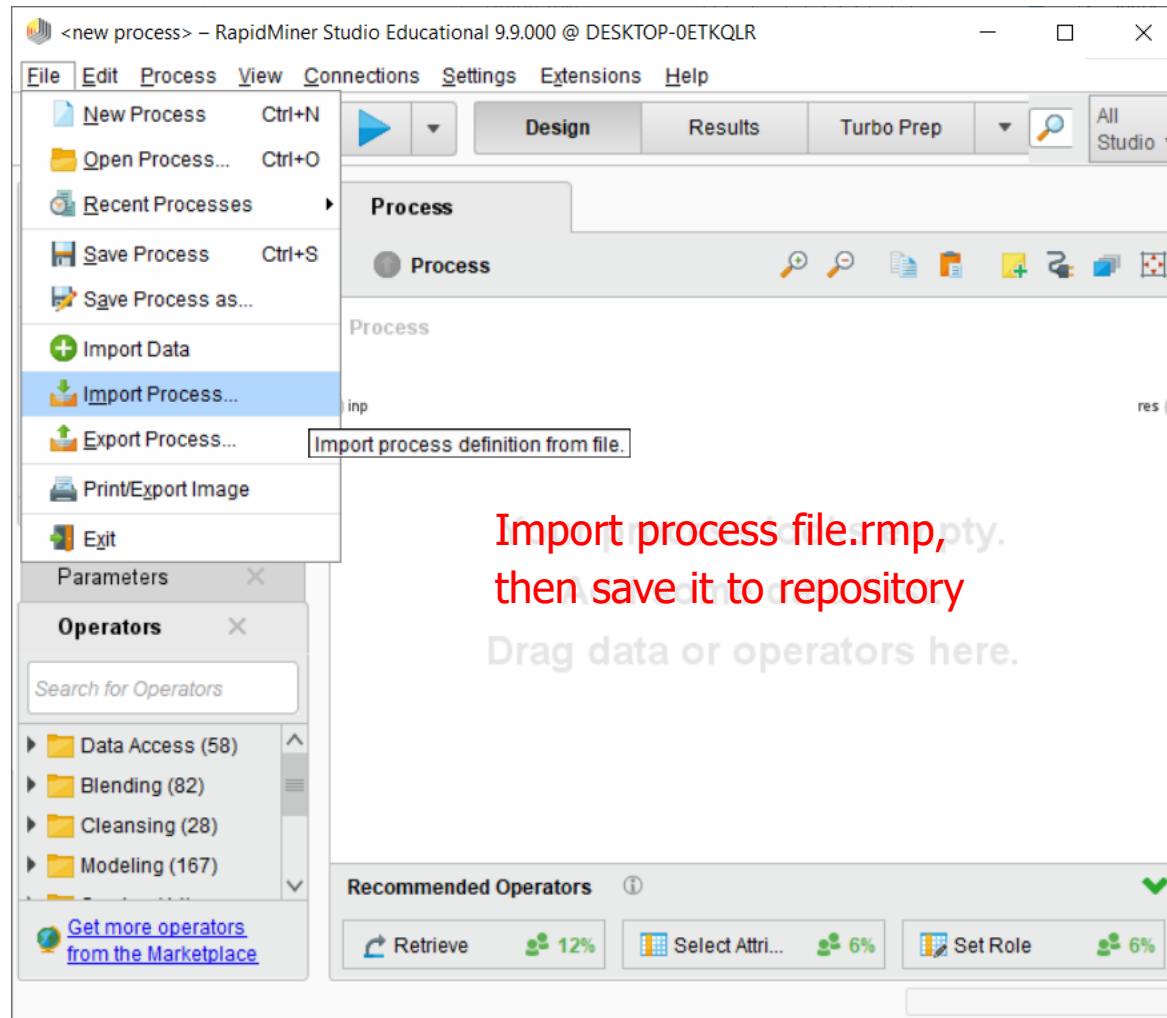
Result History

ExampleSet (Retrieve Titanic Training)

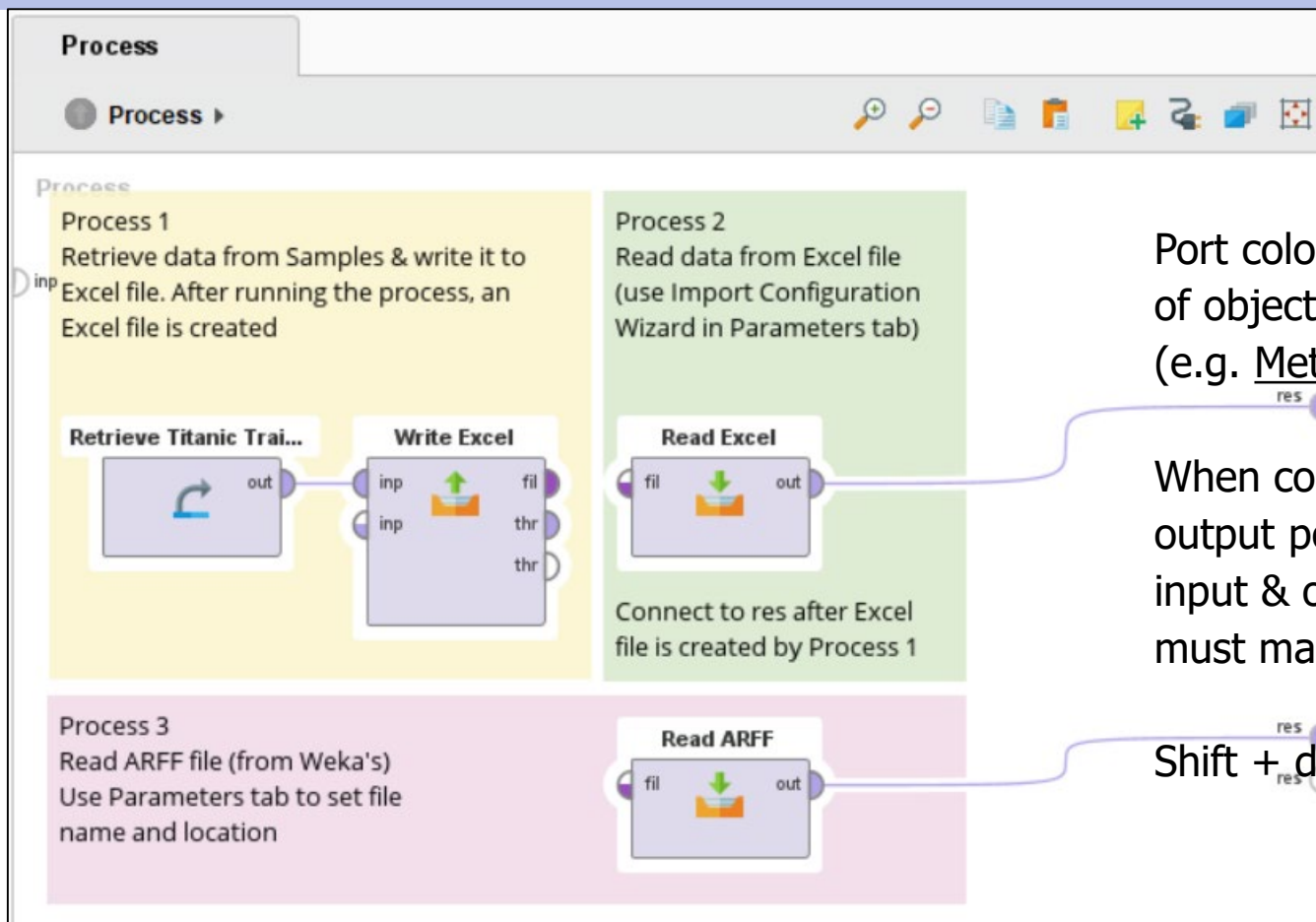
Name	Type	Missing	Statistics	Filter (7 / 7 attributes):
Label Survived	Binominal	0	 Negative Yes Positive No Values No (567), Yes (Details...)	Search for Attributes
Age	Real	0	Min 0.167 Max 80 Average 29.932	
Passenger Class	Polynomial	0	Least Second (184) Most Third (491) Values Third (491), First (241), ...[1 more]	
Sex	Binominal	0	Negative Female Positive Male Values Male (594), Female (322)	
No of Siblings or Spouses on B...	Integer	0	Min 0 Max 8 Average 0.514	
No of Parents or Children on B...	Integer	0	Min 0 Max 9 Average 0.386	
Passenger Fare				



Import process to repository



Example (2) : reading/writing files

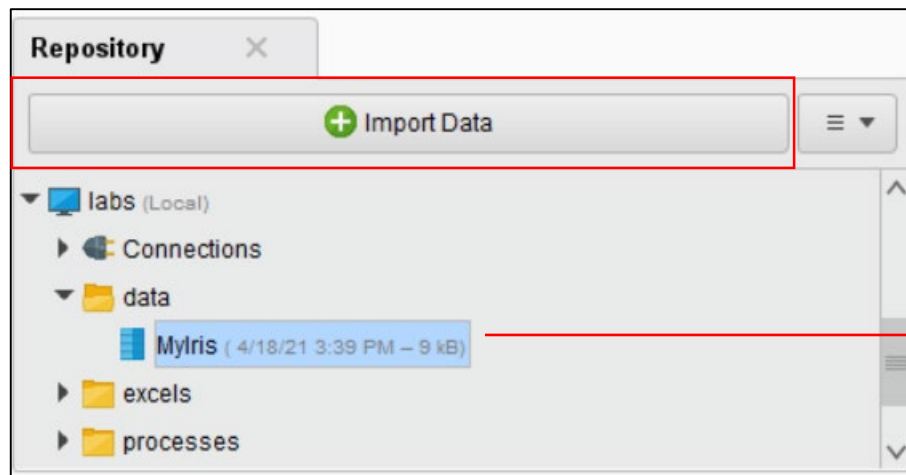


Ports

fil = file (specified in Parameters Panel)

thr = through (objects are passed through without any modification)

- ✦ When reading Excel file, RapidMiner recommends using **Import Configuration Wizard** to properly import the data
 - ▶ Otherwise, some meta data must be set manually
- ✦ After saving data to Excel file, we can import data to repository

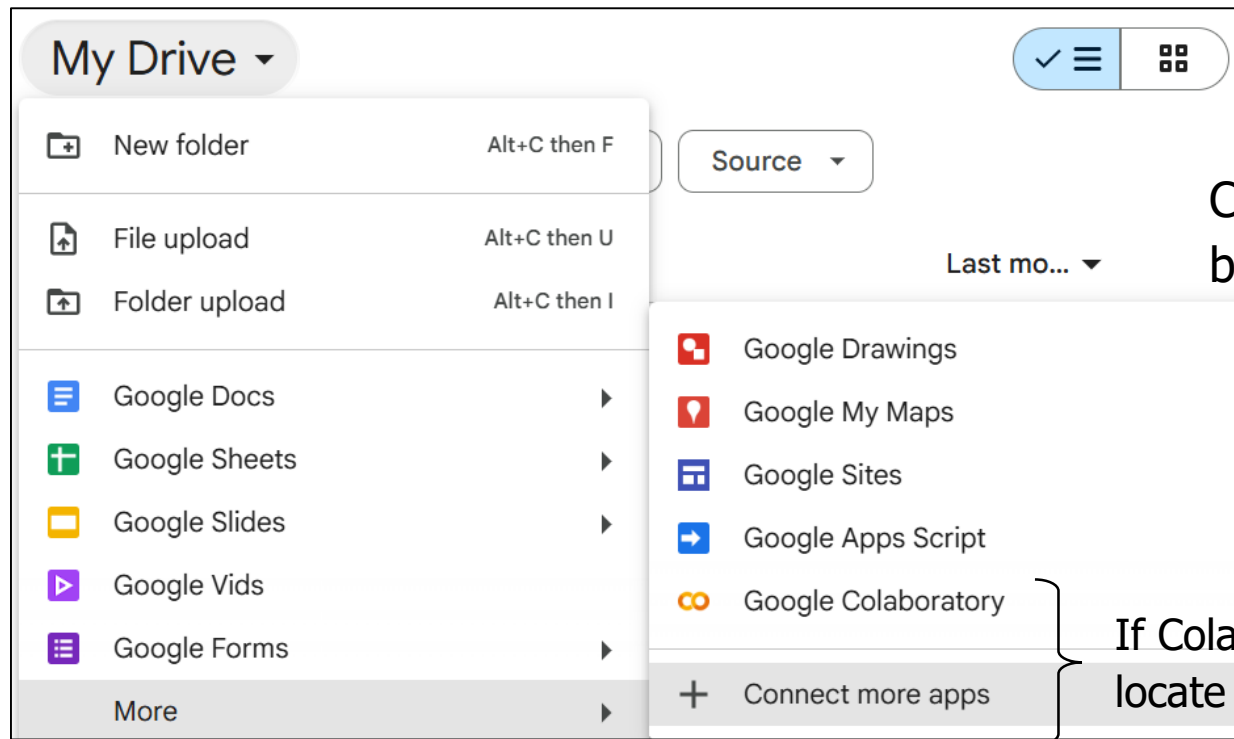


So the next time we can just drag data from repository to build a process

- ▶ Or use "Store" operator to store data (in proper format + meta data) to repository → see example in Chapter 2

JupyterLab on Cloud

✦ Google Colab (<https://colab.research.google.com/>)



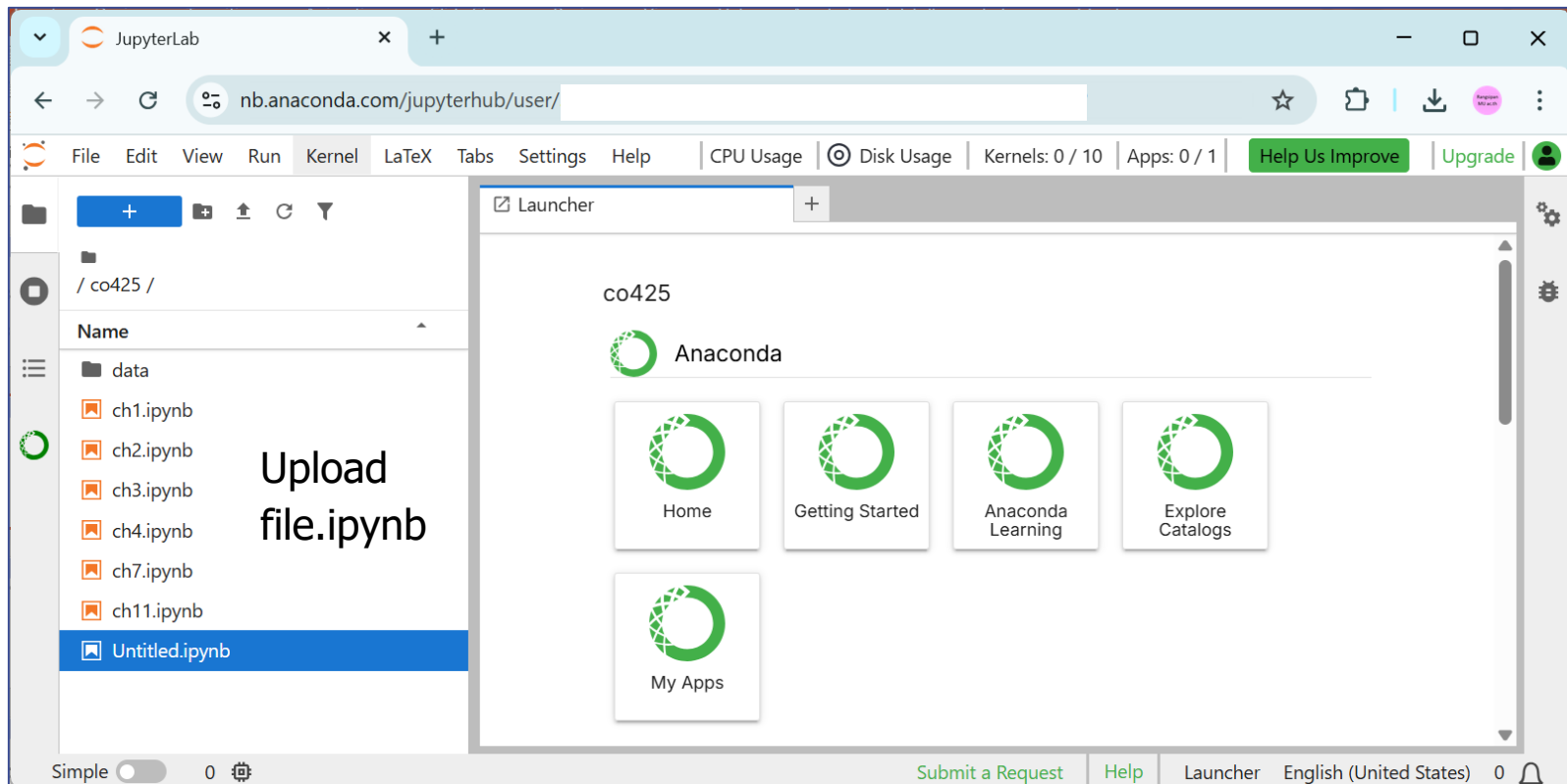
Coding suggestion
by AI (Gemini)

If Colab isn't yet available,
locate it and install

Once Colab is installed, click file.ipynb (interactive python notebook)
in Google drive to launch the notebook

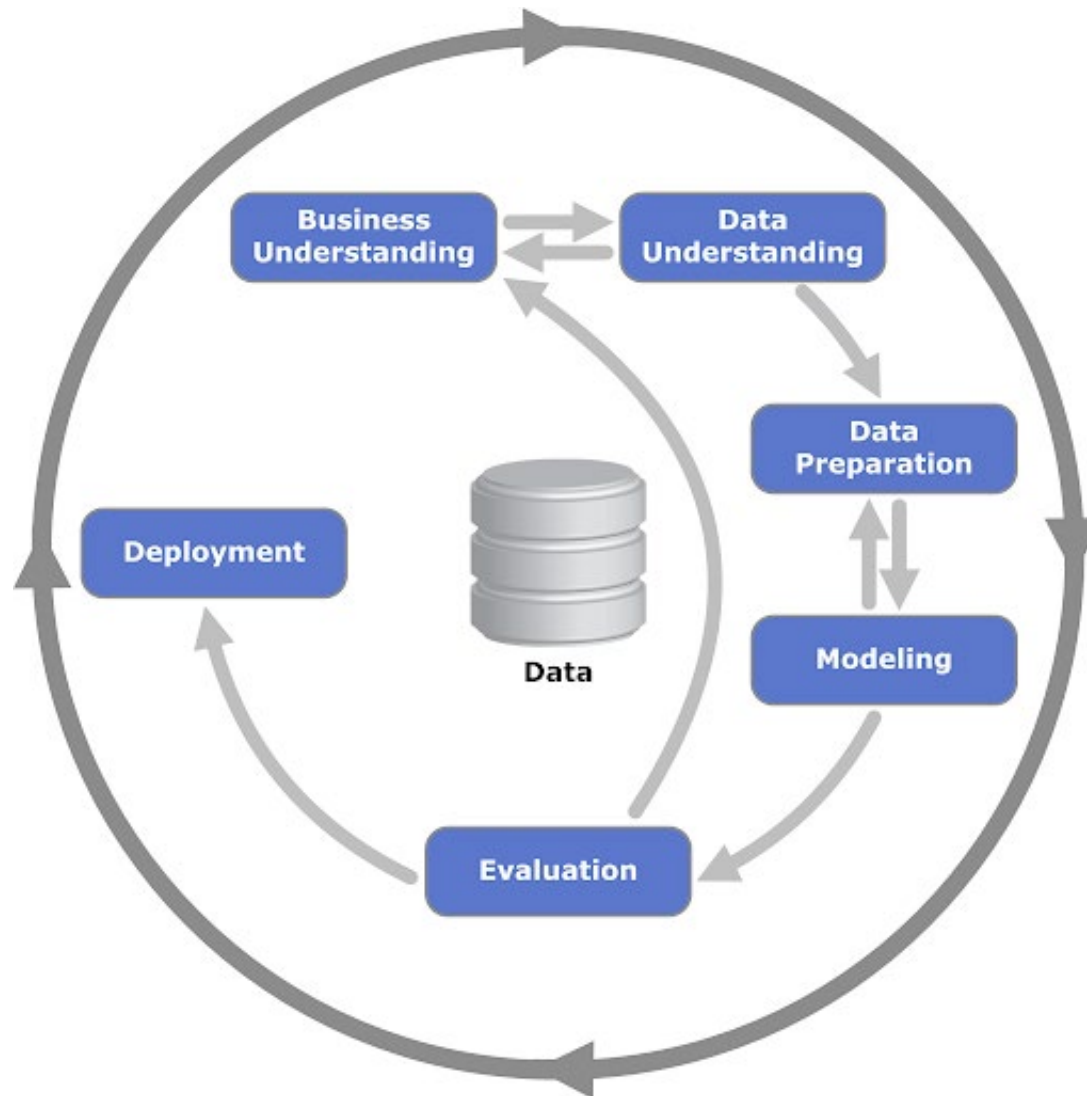
✦ Anaconda Cloud (<https://anaconda.cloud/sign-in>)

- ▶ Click "notebooks" to launch JupyterLab



CRISP-DM Framework

Cross-industry standard
process for data mining



⌘ Data preparation

- ▶ Data exploration → to get basic understanding of data and identify potential problems
- ▶ Data wrangling → format, errors, missing values, outliers
- ▶ Data transformation → appropriate data types for chosen methods
- ▶ Feature selection → relevant attributes
- ▶ Data sampling

⌘ Modeling

